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## Relatedness networks from OEC trade data

Trade data from OEC were analysed, it includes 227 different countries, 1,240 products, and two years 2022 and 2012. First of all, the 2022 year was analysed. In 2022 there remained 226 countries and 1218 products. For this year RCA, co-occurrence and proximity were calculated as in Hidalgo et al. 2007. According to the paper, a country is specialised in a product if RCA > 1, and the proximity of two products is the smaller of the two conditional probabilities that a country specialised in one is also specialised in the other.

Then for visualization of network edges were filtered using a maximum spanning tree that keeps the fewest, strongest lines that still connect every product. Initially we had 702425 edges, however after filtering only 1217 remained. After that we also added the strongest links (≥0.55), which increased the number of edges to 1642. Then each node was colored by sector, and sized by trade.

The network shows that machinery takes the middle position and located in the core, clothes form their own cluster at the side. At the same time, minerals and agriculture & food are mostly on the periphery, as well as other products.(Fig. 1)

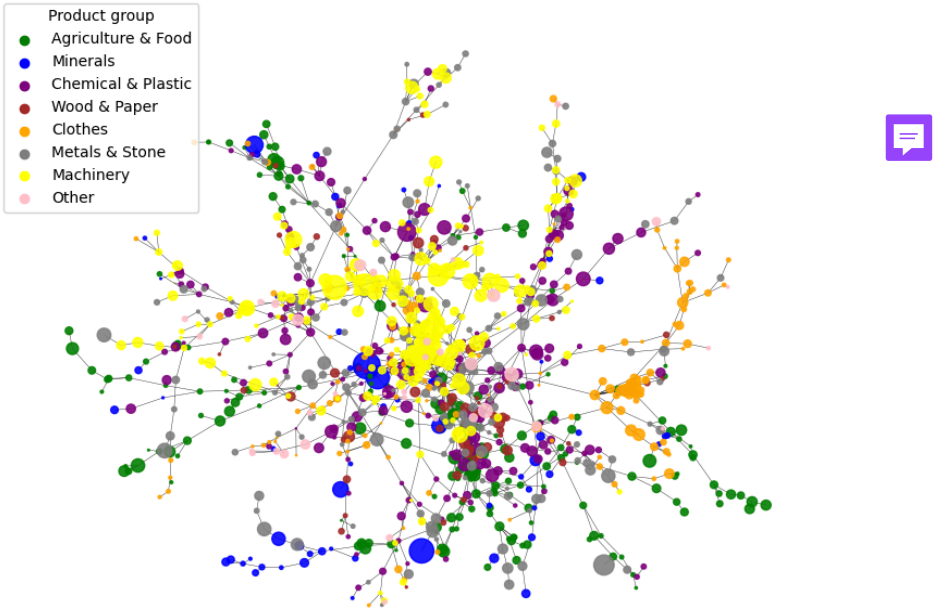



**Fig. 1. Relatedness network of products, OEC data for 2022 year. Nodes are products, edges are the maximum spanning tree and all links with proximity ≥ 0.55. Color shows the product group, node size is proportional to the square root of world exports.**

Description of network: average degree is 2.7, and the minimum is 1, because the spanning tree gives every product at least one edge, the maximum degree is 23. Products with OEC code 116..(clothes) are hubs, as they obtained the largest number of links. So most products have few links and a few hubs have many. Average weight of the edges is 0.16 and the maximum weight is 0.82. The product space is mostly weak links as 68% of pairs below 0.2 and only 0.1% of pairs pass the threshold. (Fig. 2)

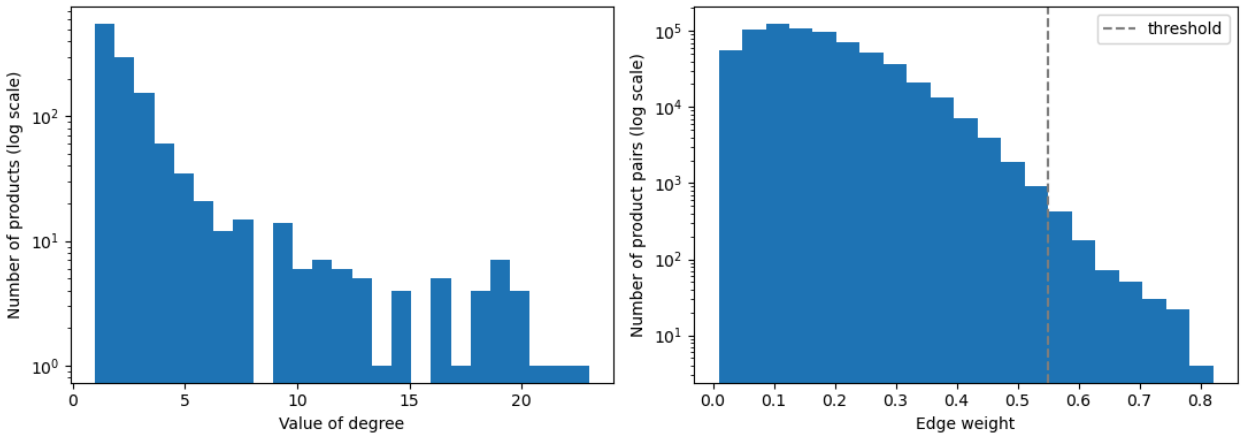



**Fig. 2. Degree and weight histograms. Left histogram: degree distribution of the filtered network. Right histogram: distribution of edge weight over all product pairs, the dashed line marks the threshold (0.55). Both y-axes are logarithmic.**

Analysing communities (clusters) it was found that 40 communities exist, using the Louvain algorithm, and the size of the largest one is 134. The largest cluster mixes Machinery (55), Metals & Stone (32), and Chemicals & Plastic (18), so they do not follow official sectors which means that products that need similar capabilities group together even when they belong to different sectors.

Core and periphery analyses showed that 18% of all products are located in the core, where core number is $\geq 2$. 24% of machinery products are located in the core, then 22% of Wood & paper and 20% of Clothes are also there. Minerals are on a periphery as its share in the core position is 0.

Then we compare two networks through the years: the 2022 network and 2012 network. The results showed that correlation of edge weights is 0.725 across common code of products. It means that product space remains stable over 10 years. Across the strong links, there were 784 of them in 2012 and 41% still persisted (above threshold) in 2022.

Scatter plot of proximities in 2012 and 2022 shows the cloud that stretches along the dashed line (that is the correlation of 0.73 made visible). However, the cloud is a little bit wide, which means that links mostly keep their strength with individual movements. On the right side of the

cloud dots lie more below the dashline which brings the conclusion that the strongest links of 2012 are a little weaker in 2022. (Fig. 3)

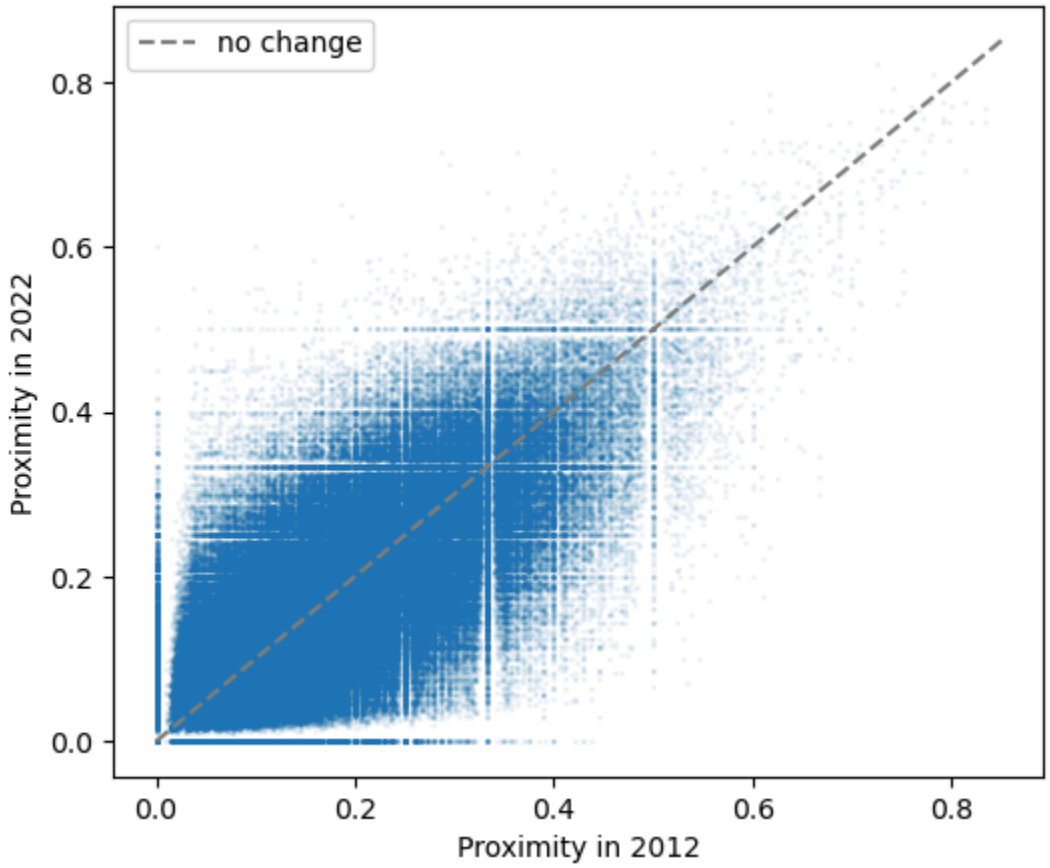



**Fig. 3, Scatter plot of proximity of each product pair in 2012 (x-axis) and 2022 (y-axis). The dashed line means no change. Correlation equals 0.73.**